Questions by topic: Scientific methodologies & sampling design - Set 1 (65 marks)

Answer all questions in the spaces provided.

0 1 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 1 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

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0 1 . 2 Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 1 . 3 Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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0 2 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	**1**	**2**	**4**	**6**	**8**	**10**	**12**	**14**
Running mean catch per night	31	18	24	21	22	22	21	22

0 2 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

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0 2 . 2 Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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0 2 . 3 Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

0 2 . 4 Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 3 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 3 . 1 Describe how they could carry out the investigation in the laboratory.

[5 marks]

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0 3 . 2 Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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0 3 . 3 Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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0 4 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 4 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 4 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 5 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 5 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 5 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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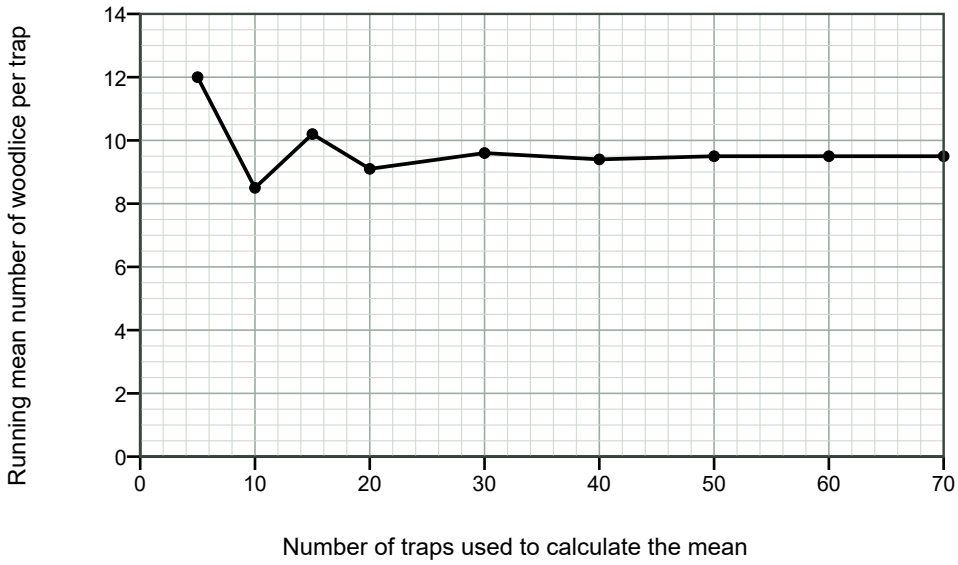
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0 6 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 6 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 6 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 6 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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0 6 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 7 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

0 7 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

0 7 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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0 7 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

0 7 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Extra space

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END OF QUESTIONS